Venn diagrams

1 How many regions are there in a Venn diagram with two overlapping sets?

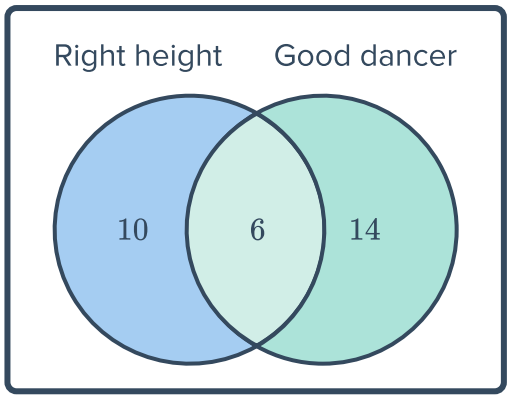
2 We are interested in the color of a card randomly drawn from a standard deck. Draw a Venn diagram to illustrates this.

3 Avril makes a Venn diagram about the possible dance partners in her dance class:

a How many students are there in Avril's class, not including Avril?

b Avril thinks her partner is suitable if they are both good dancers and the right height for her. How many suitable partners are there in the class?

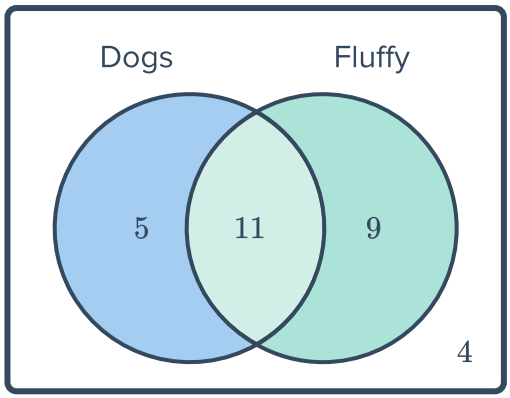
c Avril is told that her partner will be randomly assigned. Find the probability that her partner will be suitable.



4 Carl makes a Venn diagram about the possible pets he could get from the local shelter:

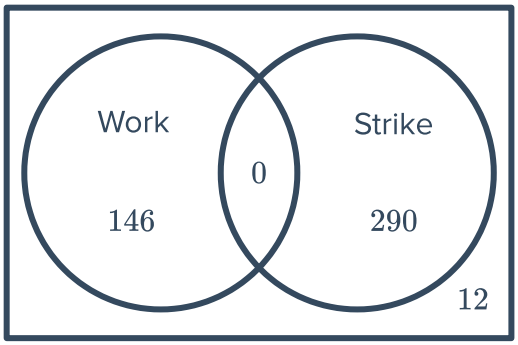
a Carl is told that a pet will be selected at random. Find the probability that his new pet will be a dog that isn't fluffy.

b Carl is told that a pet will be selected at random. Find the probability that his new pet won't be fluffy.



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5 The diagram shows the decisions of 448 workers choosing to work and workers choosing to strike on a particular day of industrial action.
Find the number of workers who chose to:



a Strike.

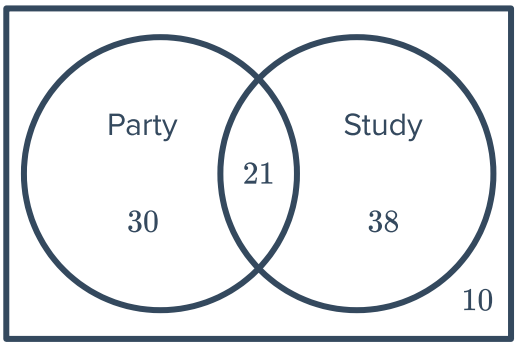
b Strike and work.

c Work and not strike.

d Work or strike.

e Neither work nor strike.

6 The Venn diagram shows the number of students choosing to study on the night before an exam, and the number of students choosing to party:
Find the number of students that chose:

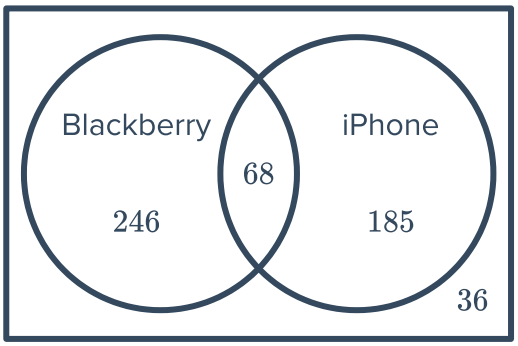


a To party.

b Not to study.

c Neither to study nor party.

7 The Venn diagram shows the decisions of 535 consumers choosing to buy an iPhone and consumers choosing to buy a Blackberry.
Find the number of consumers that chose to buy:



a A Blackberry.

b A Blackberry only.

c Both phones.

d An iPhone or a Blackberry.

e Neither phones.

f An iPhone but not a Blackberry.

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Two-way frequency tables

8 Two motels promoted packages for one night and two night stays. The following table records the results:

	One night stay	Two night stays
Motel X	463	266
Motel Y	804	817

a Find the percentage of the stays that are two night stays. Round your answer to two decimal places.

b What fraction of the two night stays was in Motel X?

c Find the percentage of the total accommodation that is in Motel X. Round your answer to two decimal places.

9 The departure timing of flights out of an airport for the months of May and June are described in the given table:

	Departed on time	Delayed
May	144	21
June	116	40

a Find the percentage of the flights in June that were delayed. Round your answer to one decimal place.

b Find the percentage of the flights in June that were on time. Round your answer to one decimal place.

c What fraction of the total number of flights during the two months were ones that departed on time in May?

10 The following table shows the number of trains arriving either on time or late at a particular station:

Day	Arriving on time	Arriving late
Monday	23	38
Tuesday	12	25
Wednesday	22	29
Thursday	30	27
Friday	11	14
Saturday	15	11
Sunday	14	10

a How many trains were late on Friday?

b How many trains passed through the station on Wednesday?

c How many trains were on time throughout the entire week?

d What fraction of trains were on time over the whole week?

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11 Katrina tossed three coins. She displayed the results in the given two-way table:

a How many times did Katrina toss heads?

b How many more times did Coin 2 land on tails than Coin 1?

c How many times did Katrina toss the coins in total?

	Heads	Tails
Coin 1	19	12
Coin 2	13	14
Coin 3	15	13

12 Michael rolled two dice. He recorded how many times each die landed on each score in the given two-way table:

a How many times did he roll a five?

b How many times did Michael roll Die 2?

c With which die did he roll more fours?

Number rolled	Die 1	Die 2
1	17	13
2	9	16
3	12	7
4	10	4
5	9	6
6	7	8

13 Thirty-six students were asked whether or not they were allergic to nuts and dairy. The two-way table is provided below:

	Allergic to nuts	Not allergic to nuts
Allergic to dairy	10	6
Not allergic to dairy	6	14

a How many students are allergic to nuts?

b How many students are allergic to nuts or dairy, or both?

c How many students are allergic to at most one of the two things?

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14 In a study, some people were asked how many times they lie in a day. 20 responders said they lie at least once a day, of which 5 were children. 15 children said they never lie, and 15 adults said they never lie.

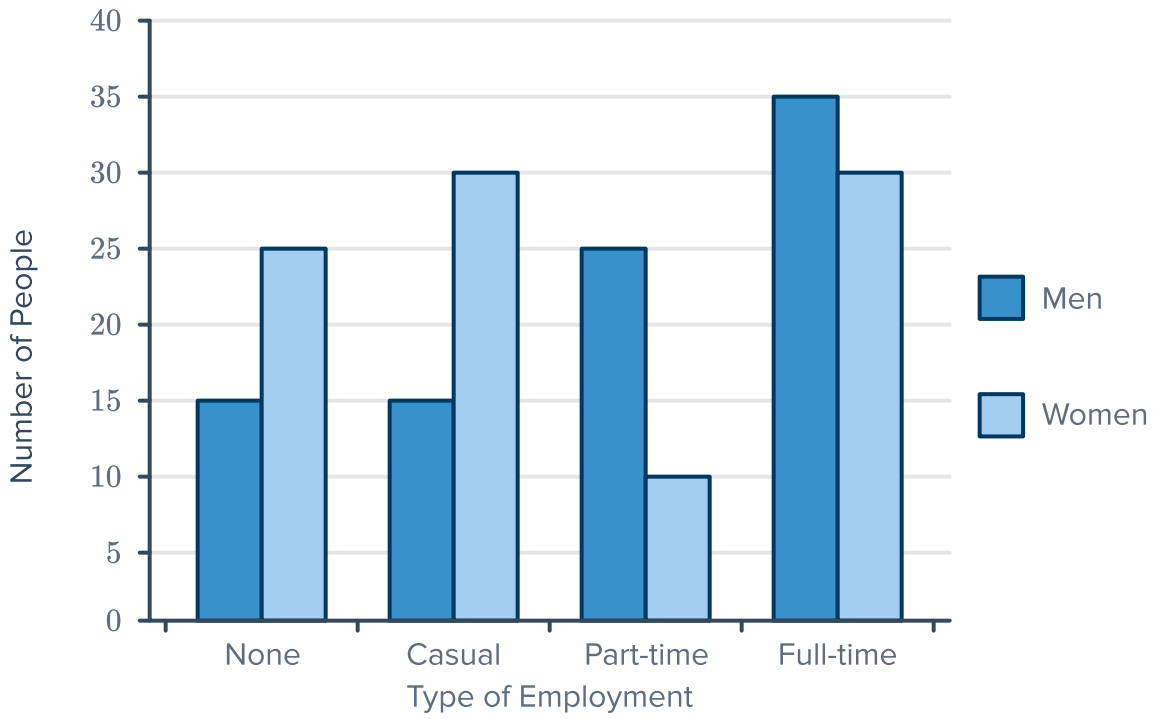
a Complete the given table.

b What proportion of responders said they never lie?

c What proportion of adults said they lie at least once per day?

	0 times	1 or more times
Children		
Adults		

15 Yuri surveyed a group of people about the type of jobs they had. He recorded the data in the following graph:



a Complete the following two-way table with the information:

	No job	Casual	Part-time	Full-time
Men				
Women				

b How many people are unemployed?

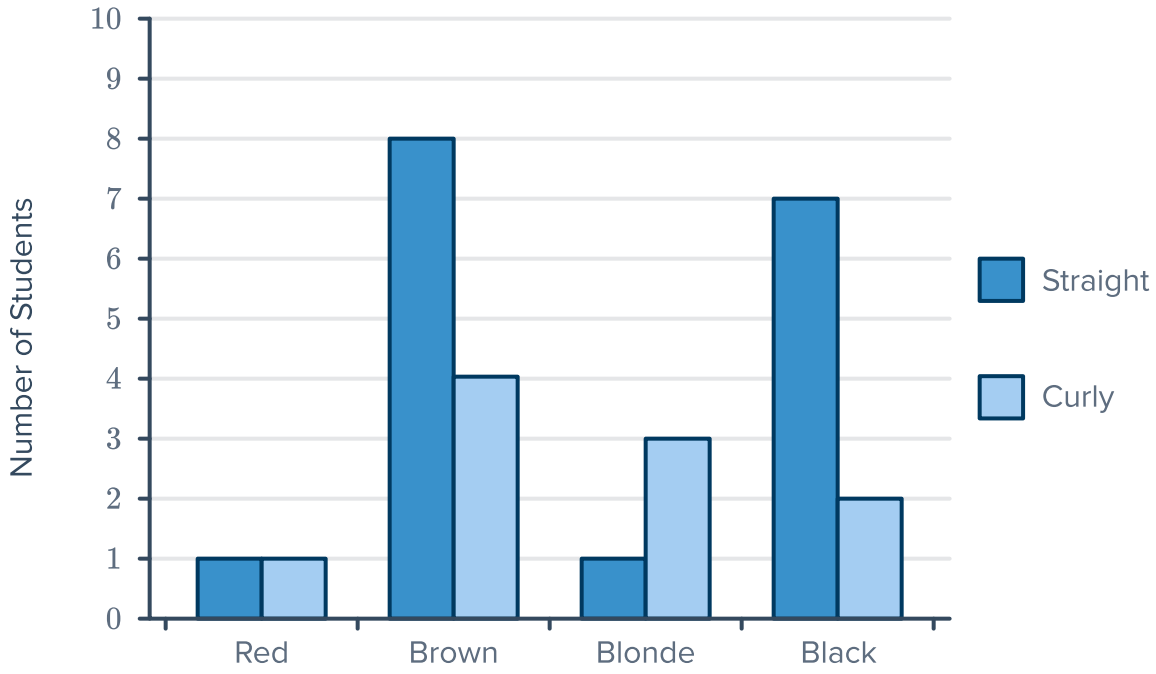
c What percentage of men are employed full-time? Round your answer to one decimal place.

d What fraction of people employed casually are male?

e If 111 people from the group are randomly selected, how many would you expect to be

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16 Miss Merryweather recorded the hair type and color of her students in the following graph:



a Complete the following two-way table with the information:

	Red	Brown	Blonde	Black
Straight				
Curly				

b Find the number of students in the class that were surveyed.

c What fraction of students have black hair?

d If 15 students from the class are randomly selected, how many would you expect to have brown hair? Round your answer correct to the nearest whole number.

17 In a biology study, the appearance of an animal when carrying a certain gene is recorded. Assume that all animals must carry Gene A or Gene B but not both. If, out of all the animals, the proportion carrying Gene A and having light fur is $\frac{4}{19}$, find the value of x.

	Gene A	Gene B
Light fur	36	37
Dark fur	28	x

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